Fraud App Detection of Google Play Store Apps Using Decision Tree

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***Abstract*— Along the rise in the various mobile applications which are used in daily life, it's more necessary than ever to stay on top of things to decide which are safe and which don't. It is impossible to pass judgment. Our system is based on four parameter that include ratings, reviews, in app purchases and Contains ad to predict. Our system compares three models Decision Tree classifier, Logistic Regression and Naïve Bayes. These models were further analyzed on four parameters of F1 score, Recall, Precision and Accuracy. A good F1 score should be greater than 0.7 and a recall score greater than 0.5 is considered to be good with higher precision and accuracy. On analysis we found Decision tree model as a good model with accuracy of 85%, F1score of 0.815, Recall value of 0.85 and precision of 0.87**

Keywords: Decision Tree, Google Play Store Apps, Reviews, Ratings, In app purchases, Contains Ad

I. INTRODUCTION

The use of mobile phones has increased as technology advances. The development of different play store apps on various major platforms, like popular Android and iOS, has exploded. It has become a major problem in the business intelligence area due to its rapid daily expansion through its daily use, marketing, and development. The market is becoming more competitive as a result [7]. Companies and software developers are in intense competition with each other to prove their product quality and invest a lot of time and effort in hiring clients to maintain their future success.

Customer ratings, and updates on each program they have the opportunity to download play a very important role. This can be a strategy for engineers to identify their flaws and integrate them into the design of a new product that meets the needs of the people. [5].

As a general practice, instead of relying on traditional marketing strategies, under the trees App developers have the ability to majorly support their apps, and ultimately, manage the rank in the App Store. This is often accomplished by using so-called "bot ranch" or "water army" to increase the number of downloads and audits [15].

Occasionally, for the benefit of the developers, they will hire groups of people who will collectively commit fraud and offer spamming comments and ratings on an application. Crowd turfing is the word for this type of behaviour.

As a result, it is critical to ensure that consumers are presented with accurate and authentic feedback prior to installing an app in order to minimize errors. In order to overcome and analyze the many comments and ratings that are supplied for each application, an automated method is required [6].

Because mobile phones are such a common requirement, it is critical that suspicious program be designated as fraud so that play store users may uniquely identify them. It will be impossible for the user to tell if the remarks or ratings that they scroll over are a scam or genuine for their advantage. By offering a comprehensive perspective of ranking fraud detection system, we propose a system that would identify such fake applications on the Play or App store [2].

We can acquire the probability of determining whether an app is fake or not, therefore we present a system that uses four features that are In app purchases, Contains Ad ,ratings and reviews to determine the probability of an app whether it's scamming its consumers or not[5].

We start the system by taking into account these four features that are the most crucial in determining the target. Then based on these features the collected data through scraping is being trained through various classification models and then selecting the right and accurate model for the system. In this stage of the selection we got various models with different accuracy Naive Bayes with 83%, Logistic Regression with 84% and Decision tree with 85% accuracy [2].

II. LITERATURE REVIEW

Nevon Projects has proposed such a comprehensive framework that can be expanded with additional evidence generated by the domain to detect quality fraud. It is one of the most advanced projects to detect fraudulent applications using information algorithms. This program provides only 75-80% accuracy in detecting fraudulent applications.

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On paper [7], they provided a complete overview of the facts and a recommended fraud detection framework. They tested three types of verification: Quality-based guarantees, rating-based guarantees, review-based validation.

In [2] paper, they view only updates as parameters with algorithm naïve bayes. They have developed a system that involves detecting fraudulent applications through emotional commenting and data processing.

In [3] paper, first take a look at the app based totally on analytics evaluation as a way to decide whether the software is fake or proper. When enhancing your e-mail, Use to test spelling of the file. through the usage of a simple facts set of rules and a manage analyst that exposes app fraud and once more primarily based on feedback from users we provide key-word analysis and based totally on scale. in this manner we also assist with what the consumer thinks approximately our app. on this fraud detection utility, the administrator also offers an app hyperlink that ought to be delivered to the software in which the person when logged in can see the app details and spot the hyperlink for that software. For us in which managers upload the app and offer a hyperlink for that from the Apple keep and Google Play store. to improve our app we are able to ask the consumer after the usage of our app to get comments, we use this sort of remarks from the consumer and help us enhance the app where the consumer the use of our app will at once see all apps analyzed via the administrator and introduced by way of admin. Consequently, the user receives all the records approximately that app and whether the app is fake or not. While an administrator pronounces fraud rather than their fraud, that saves the consumer time and offers person safety.

The JP INFOTECH project is an effective algorithm for identifying the best times for each application based on record level. By analyzing the operating environment of applications, this system finds that Fraud Apps typically has different levels for each of the leading session patterns compared to standard applications. Another fraudulent proposal points to historical records of the level of applications that lead to the development of three jobs in order to always obtain a level based on a fake topic. In addition, two types of fraudulent proposals are proposed based on application rating and review history.

On paper [4], they consider only the risk signals required by the application using the risk summary. They have developed a system where the required features of risk signals and a limited number of risks for android apps remember the ultimate goal

On paper [1], which suggests a way to detect fraud fee the use of IP address detection. The use of a cellular user IP deal with has also been one of the maximum latest emails surveyed. In the advert for a cellular app, an app called a simply perverted app is developing. Today, reputation and expectations play an important function within the cell marketplace. Experiments accumulate to provide a position in every application. However IP snooping permits users to exchange their IP address and permits them to price the utility more than once.

In the paper [6], the creators have explored the issue of detecting shilling attacks one and a half times in measuring data. Dependent philosophy can be applied to the proposition of credible material and learning that is less controlled.

TABLE 1

S. No.	Title of Papers	Year	Sensors and Technology used
1	Detection od fraud apps using sentiment analysis	2020	Sentiment Analysis
2	Detection of Fraud Apps using Sentiment Analysis	2018	Sentiment Analysis
3	Application of Machine Learning on Fraud App Detection	2021	Machine Learning
4	Detecting Fraud Apps using Sentiment Research	2019	Sentiment Research
5	Fraud App Detection using Fuzzy Logic Model Based on Sentiment of Reviews	2018	Fuzzy Logic
6	Fraud Application Detection Using Data Mining Techniques	2017	Data Mining
7	Detection of Fraud & Malware Apps in Google Play	2018	Evidence Aggregation

III. METHODOLOGY

Steps:

1. *Data Collection*

- Collecting dataset of 5500 apps from Kaggle and fraud apps from various articles.

Kaggle [7] is a tremendous platform in which it is easy for every user to discover and post datasets, check and construct models with other information. Scientists and Machine learning engineers solve data challenges that require information via the usage of statistics from Kaggle. Kaggle provides a no-setup, custom-made, Jupyter Notebooks environment. It also provides access free to GPUs and a massive repository of various community published data & code.

2. *Data Pre-processing and filtration*

- Remove missing value, only required columns are kept. This is done through pandas.

Pandas [7] is used especially for facts evaluation. Panda imports statistics from diverse document formats including comma-separated values, JSON, square, and MS Excel. Pandas allows for a filtration of facts together with merging,

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reshaping, sorting, and data processing, in addition to facts conflicts.

3. Model creation and training

- In the model we have taken 75% of the data for training and rest 25% for testing.We have Made model of decision tree, naive bayes algorithm and logistics regression. Further after comparing all the models, the accuracy of decision tree was highest.
- Compared accuracy of Naive Bayes (83%), Logistic Regression (84%) and Decision Tree (85%) algorithm.

a. Naïve Bayes Algorithm

Naïve Bayes algorithm [7] based on the Bayes theorem, a supervised learning algorithm and used to solve classification problems. It is especially used in classifying text that includes high-quality training databases. Its Classifier is one of the simplest and most efficient programming algorithms that helps to build faster machine learning models that can make faster predictions. It is a process of possible separation that is it predicts the outcome based on the potential of the object. Other common examples of the Naïve Bayes Algorithm include Sentimental analysis, spam filtering and article classification. It is therefore used as the most popular alternative to text-sharing problems.

b. Logistic Regression

Logistic regression [6], apart from its name, it is a classification model even after having regression in the name. Logistic regression is defined as a process of modeling the probability of a distinct outcome when given an input variable. It is a type of classification model, and is very easy to emphasize and it also achieves really good execution with linearly distinct classes. This kind of analysis can really help in predicting the likelihood of an event being happening or a choice being made. It is a considerably used algorithm for classification in the industry. Logistic regression model is a statistical methodology for binary classification which can be generalized to a multiclass classification. It is a considerably useful analysis method for various classification problems, determining whether a new sample fits better into a category.

c. Decision Tree Algorithm

Decision Trees [6] is a supervised algorithm which is used for both classification as well as regression. The purpose of the decision tree is to create a model which predicts the value of target variable with the help of learning simple decision rules gathered from the various data features. The Decision tree can be learned by source set being split into various subsets which are based on feature value test. The following process is being repeated on each and every derived subset in a manner which is recursive and is known as *recursive partitioning*. This recursion is finished whenever the subset at a particular node has the same value as that of the target variable, or when no longer splitting adds value to the predictions. The building of a decision tree classifier technique does not need any domain knowledge or parameter setting, and because of this it is appropriate for probing knowledge discovery. High dimensional data can be handled well by decision trees. In broad terms the decision tree classifier is considered to have a good accuracy.

4. Deploying of model

- Made user interface web page using flask. In which the user puts the url in the search box and gets the result as "fake" or "safe to use" for the app.

Flask is a web framework, because of this flask offers you with gear, libraries and technology that allow you to build a web utility.

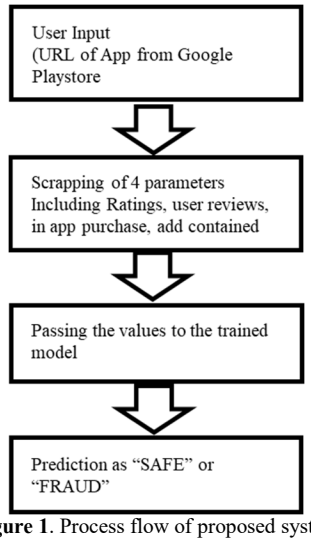


Figure 1. Process flow of proposed system

IV. RESULTS

The sole purpose of the given proposed system is majorly to review the fraud detection of Google play store applications and to use the four parameter methods to differentiate certain fraudulent applications or commonly referred to as spam applications. Experimental analysis is performed on different types of methodology in the proposed manner for the detection of fraud or fake applications. Our system will receive fraud with four types of evidence, such as ad-based ratings, in-app purchases and evidence-based reviews. In addition, the development-based integration approach incorporates all four aspects to detect fraud. Various machine learning model were implemented which provided different results for the accuracy. By analysis, we found that our given proposed method provides 85% accuracy compared to other algorithms. While independent thinking still exists, the Decision Tree section performs better compared to other models such as the recession and the Naïve Bayes. It is an intuitive algorithm for separation problems. It is a reliable real-time guess, a setback problem. Decision trees can manage non-linear data sets effectively. It plays a role in decision-making in various fields of life, including engineering, social planning, business, and even law.

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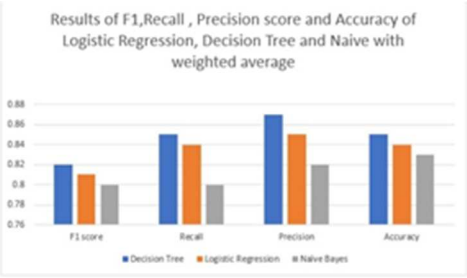


Figure 2. Results Graph

The Results of F1, Recall, Precision and Accuracy Scores of Logistic Regression, Decision Tree and Naïve with weighted average.

TABLE 2. COMPARATIVE ANALYSIS ON DIFFERENT MACHINE LEARNING ALGORITHMS FRAUD APP DETECTION

	F1	Recall	Precision	Accuracy
Decision Tree	0.815	0.85	0.87	0.85
Logistic Regression	0.81	0.84	0.85	0.84
Naïve Bayes	0.8	0.83	0.82	0.83

V. CONCLUSION

In the Era of growing technology, the threat to security is also becoming a major issue and a part of it, today we have a number of apps listed on the Google app stores which also includes various fraud apps that are threat to users' privacy and data.

In our model we have worked precisely to detect fraudulent software using 4 parameters including scales, review scores, in-app purchases and content additions. We compared the accuracy of the three algorithms when the resolution tree appeared to be 85% higher. The framework is measurable and can be expanded to further proven domain-based fraudulent evidence. Demonstrated the effectiveness of the proposed system, algorithm detection measurement and standardization of level fraud operations. This can be used effectively to rate fraudulent play store application in play store for security purposes.

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